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Introduction

With the rapid development of Internet of Things (IoT) applications, smart safety systems have become an important part of modern technology. This project focuses on designing and implementing a Smart Car Collision Warning System that helps prevent accidents by warning the driver when the car is approaching a wall or any obstacle. The system relies on an Ultrasonic Sensor to measure the distance between the car and the obstacle, then provides visual and sound alerts to the driver.

Project Description

The Smart Car Collision Warning System is an electronic circuit designed to enhance safety during driving or parking. The system continuously measures the distance between the vehicle and nearby objects. When the distance decreases, warning signals are activated to alert the driver. The warning process is done gradually using LED indicators and a Buzzer, which makes the system easy to understand and effective in real-life situations.

Experiment Description

In this experiment, Arduino Uno is used as the main control unit of the system. The Ultrasonic Sensor (HC-SR04) sends ultrasonic waves toward an obstacle and receives the reflected waves. The time taken for the waves to return is used to calculate the distance. Depending on the measured distance, LEDs turn on gradually as the car gets closer to the obstacle, and when the distance reaches a dangerous level, the buzzer is activated to produce an audible warning. This method allows the driver to react early and avoid collision.

Components Used

The components were used to build the circuit: Arduino Uno, Ultrasonic Sensor (HC-SR04), Buzzer, 6 LEDs, 6 Resistors (200 Ohms), Breadboard, and Connecting Wires.

Circuit Connection

The circuit is connected in such a way that the Ultrasonic Sensor is connected to Arduino digital pins for distance measurement. LEDs are connected through resistors to indicate distance levels, and the Buzzer is connected to provide an audio warning when

the distance becomes critical. A circuit connection diagram or simulation image should be added here.

Code used

```
int ledPins[6] = {2, 3, 4, 5, 6, 7};
int numberOfLeds = 6;

int trigPin = 10;
int echoPin = 11;
int buzzerPin = 9;
// buzzer + pin 9 , - GND , and add 220 Ohms
int travelTime;
int distance;

void setup() {
  for (int i = 0; i < numberOfLeds; i++) {
    pinMode(ledPins[i], OUTPUT);
  }

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
}

void loop() {

  sendSoundWave();
  travelTime = pulseIn(echoPin, HIGH);
  distance = travelTime * 0.034 / 2;

  if (distance <= 20) {

    int n = map(distance, 0, 20, numberOfLeds, 0);

    for (int i = 0; i < numberOfLeds; i++) {
      if (i < n) {
        digitalWrite(ledPins[i], HIGH);
      } else {
        digitalWrite(ledPins[i], LOW);
      }
    }

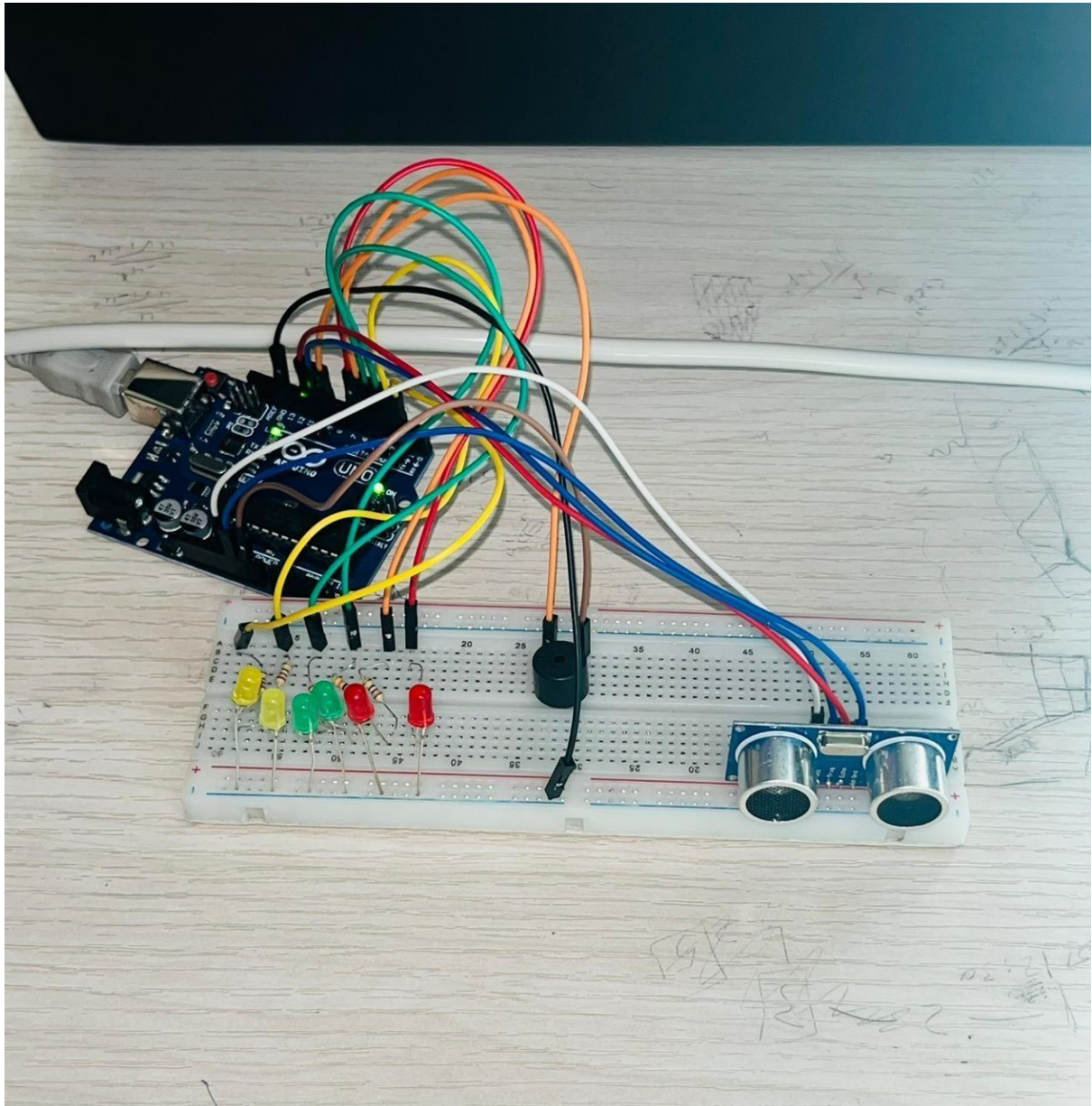
    if (distance <= 3) {
      digitalWrite(buzzerPin, HIGH);
    } else {
```

```
        digitalWrite(buzzerPin, LOW);
    }
} else {
    for (int i = 0; i < numberOfLeds; i++) {
        digitalWrite(ledPins[i], LOW);
    }
    digitalWrite(buzzerPin, LOW);
}

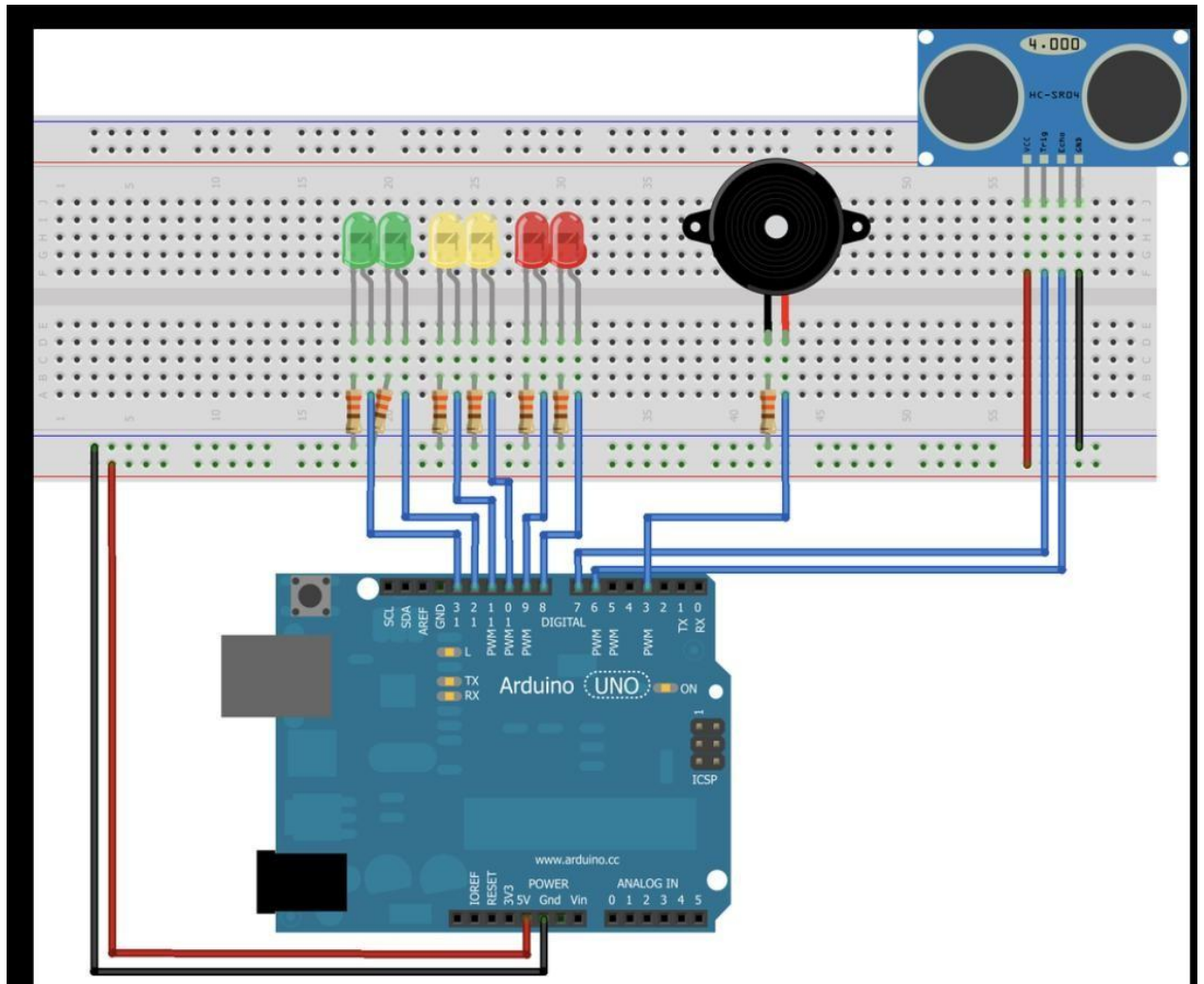
delay(10);
}

void sendSoundWave() {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
}
```

Real Circuit Image



Simulation Connection Diagram



Conclusion

The Smart Car Collision Warning System demonstrates a practical application of IoT concepts in safety systems. It provides an efficient and low-cost solution to reduce collision risks by using ultrasonic sensing and real-time alerts. The project can be further improved by integrating wireless communication or mobile notifications in future work.